



# lichen planus

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INTERN

# Introduction:

- ▶ Lichen planus is a disease of the skin and/or mucous membrane that resembles lichens.
- ▶ It is thought to be the result of an autoimmune process with an unknown initial trigger.
- ▶ Where the trigger is known a lesion as a lichenoid lesion.





# Epidemiology :

- ❑ **Risk** for the condition include:
  - ✓ exposure to medicines, dyes, and other chemicals (including gold, antibiotics, arsenic, diuretics, chloroquine)
  - ✓ Disease such as hepatitis C
- ❑ **Race**: no racial predispositions.
- ❑ **Sex**: lichen planus affects women more compared to men (3:2) ratio.
- ❑ **Age**: more than two third of LP patients are aged 30-60 yrs however it can occur at any age.

# Lichen planus

- ▶ Lichen planus is an uncommon, chronic, pruritic disease characterized by violaceous flat-topped papules that are usually seen on the wrists and the legs
- ▶ Characterized by : 9 P's
  - Papulosquamous disorder
  - Pruritic
  - Polyangular with
  - Plain topped
  - Pigmented
  - Purple coloured
  - Papules and plaques
  - Pterygium unguium present in nails
  - Penile annular lesions



- ▶ It is a cell mediated immune response of unknown origin.
- ▶ It may be found with other disease of altered immunity, such as ulcerative colitis, lichen sclerosis, myasthenia gravis, etc
- ▶ Lichen planus has been found to be associated with hepatitis C virus infection ,chronic active hepatitis and primary biliary cirrhosis.
- ▶ It is most likely an immunologically mediated reaction, through the pathophysiology is unclear.



# Sign and symptoms:

- ❖ The following may be noted in the patient history:
- ✓ Lesions initially developing on flexural surfaces of the limbs, with a generalized eruption developing after a week or more and maximal spreading within 2-16 weeks .
- ✓ Pruritus of varying severity, depending on the type of lesion and the extent of involvement
- ✓ Oral lesions that may be asymptomatic, burning, or even painful
- ✓ In cutaneous disease, lesions typically resolve within 6 months (>50%) to 18 months (85%); chronic disease is more likely oral lichen planus or with large, annular, hypertrophic lesions and mucous membrane involvement



❑ In addition to the widespread cutaneous eruption, lichen planus can involve the following structures:

a. Mucous membranes

b. Genitalia

c. Nails

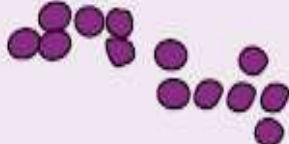
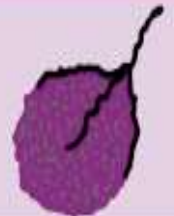
d. Scalp

# Clinical presentation:

Lichen planus has several variations, as follows:

- ▶ *Hypertrophic lichen planus*
- ▶ *Atrophic lichen planus*
- ▶ *Erosive/ulcerative lichen planus*
- ▶ *Follicular lichen planus (lichen planopilaris)*
- ▶ *Annular lichen planus*
- ▶ *Linear lichen planus*
- ▶ *Vesicular and bullous lichen planus*
- ▶ *Actinic lichen planus*
- ▶ *Lichen planus pigmentosus*
- ▶ *Lichen planus pemphigoides*



| <i>Variant</i>                      | <i>Change</i>                                                                       | <i>Morphology</i>                                                      | <i>Sites of predilection</i>                           |
|-------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Annular</b><br>(Fig. 4.21A)      |    | Hyperpigmented flat center<br>Violaceous elevated periphery            | Face, glans penis                                      |
| <b>Actinic</b><br>(Fig. 4.21B)      |    | Annular lesions with thready edge<br>Perilesional hypopigmented halo   | Face, dorsal/dorsolateral aspect of upper extremities. |
| <b>Linear</b><br>(Fig. 4.21C)       |    | Papules arranged linearly                                              | Extremities                                            |
| <b>Follicular</b><br>(Fig. 4.21D)   |    | Perifollicular violaceous papules leading to cicatricial alopecia      | Scalp, trunk, medial aspect of extremities             |
| <b>Hypertrophic</b><br>(Fig. 4.21E) |   | Central depigmentation<br>Verrucous hyperkeratotic papules and nodules | Shins                                                  |
| <b>Bullous</b><br>(Rare)            |  | Lesion of LP<br>Bulla                                                  | Extremities                                            |

# CHARACTERISTICS:

- ▶ **Primary lesion**: Flat-topped, violaceous papules and papulosquamous lesions appear.

On close examination of a papule, preferably after the lesion has been wet with an alcohol swipe, intersecting small white lines or papules (**Wickham's striae**) can be seen. These confirm the diagnosis.

- ▶ Uncommonly, the lesions may assume a ring-shaped configuration (especially on the penis) or may be hypertrophic (especially pretibial), atrophic, or bullous.
- ▶ On the mucous membranes, the lesions appear as a whitish, lacy network



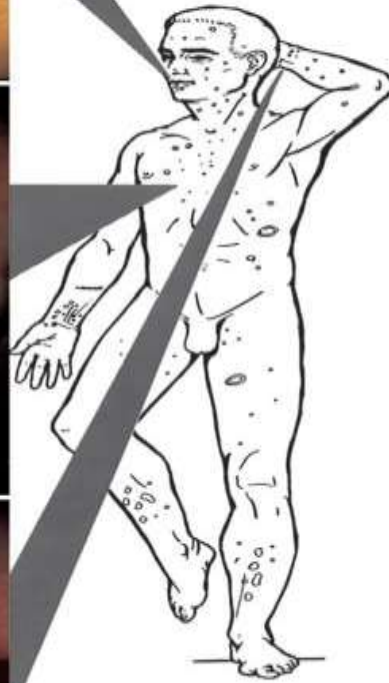
- ▶ **Secondary lesion:** Excoriations and, on the legs, thick, scaly, lichenified patches have been noted. Lesions are often rubbed rather than scratched because scratching is painful.

- ▶ **Distribution:**

Most commonly, the lesions appear on the flexural aspects of the wrists and the ankles, the penis, and the oral mucous membranes, but they can be anywhere on the body or become generalized.

Lesion may appear at the sites of trauma (koebner's phenomenon)





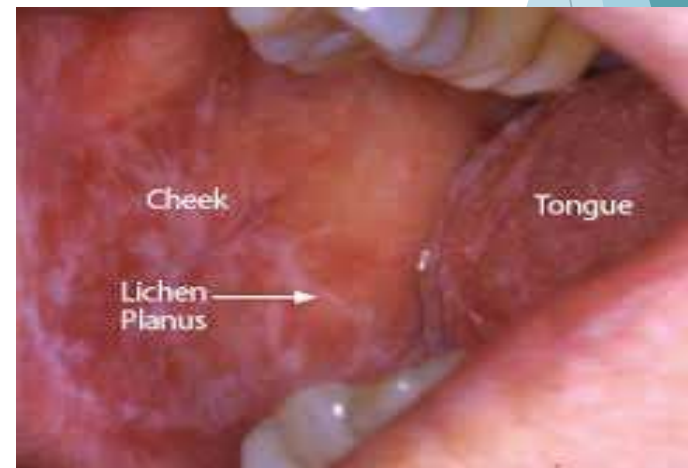




# Clinical Forms:

## ❖ Oral mucosa:

- ❑ Involved in 50% of patients with cutaneous LP and may be sole manifestation of LP in 10% of patients.
- ❑ **Reticular** the most common presentation of oral lichen planus, characterized by the net like appearance of lacy white lines, oral variant of **Wickham's striae**. This is usually asymptomatic.
- ❑ **Erosive/ulcerative** characterized by oral ulcer presenting with Persistent irregular areas of redness, ulceration and erosion covered with a yellow slough, the gums are involved described as **desquamative gingivitis**.



## ❑ Genital mucosa:

- ▶ Involved in 25% of patients
- ▶ Annular lesions on the glans penis.



## ❑ Scalp lesions:

- ▶ Follicular lesions of LP on the scalp subside with scarring and result in cicatricial (scarring) alopecia





## ❑ Nail changes:

- ▶ Seen in 15% of patients (less frequently in children).

Common manifestations of LP of nails are:

- ✓ Thinning and distal splitting of nail plates.
- ✓ Longitudinal ridging.
- ✓ Tenting of nail plate (pup tent sign).
- ✓ *Trachyonychia*: characterized by nail roughness due to excessive longitudinal ridging(sand paper nails).

**Pterygium formation is diagnostic.**



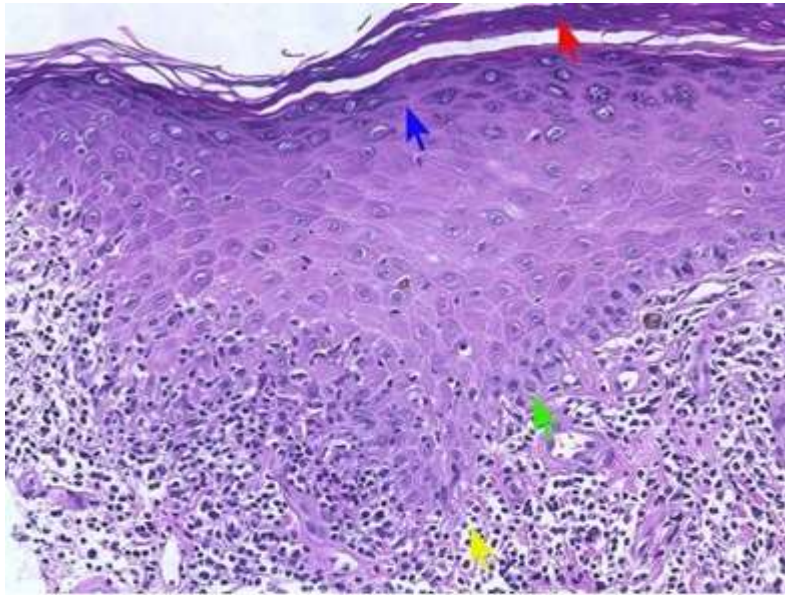
# Diagnosis:

- ▶ Direct immunofluorescence study reveals globular deposits of immunoglobulin M (IgM) and complement mixed with apoptotic keratinocytes.
- ▶ No imaging studies are necessary.
- ▶ Histopathologically

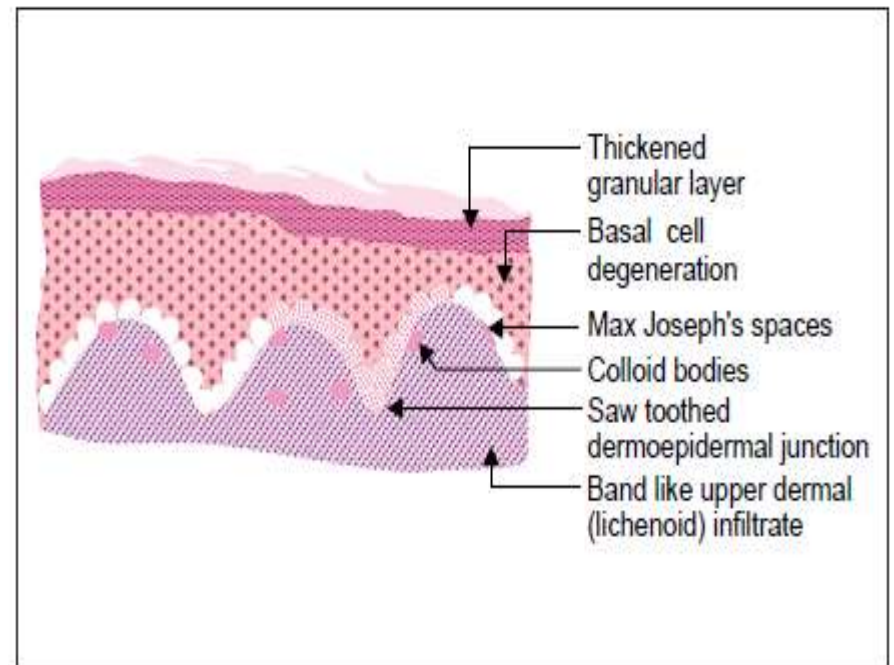
# Histopathology:

Distinguishing histopathologic features of lichen planus include the following:

- ▶ Hyperkeratotic epidermis with irregular acanthosis and focal thickening in the granular layer
- ▶ Degenerative keratinocytes (colloid or Civatte bodies) in the lower epidermis; in addition to apoptotic keratinocytes, colloid bodies are composed of globular deposits of IgM (occasionally immunoglobulin G [IgG] or immunoglobulin A [IgA]) and complement
- ▶ Linear or shaggy deposits of fibrin and fibrinogen in the basement membrane zone
- ▶ In the upper dermis, a bandlike infiltrate of lymphocytic (primarily helper T) and histiocytic cells with many Langerhans cells



-  **Hyperkeratosis**
-  **Thickened granular layer**
-  **Jagged outline of epidermis**
-  **Lymphocytes obscuring the dermal-epidermal infiltrate**



# Management:

- ▶ Lichen planus is a self-limited disease that usually resolves within 8-12 months.
- ▶ Mild cases can be treated with fluorinated topical steroids.
- ▶ More severe cases, especially those with scalp, nail, and mucous membrane involvement, may necessitate more intensive therapy.



# Pharmacological management:

## ► Cutaneous lichen planus:

1. Topical steroids (first-line treatment); systemic steroids;
2. Oral regimens like metronidazole, acitretin, methotrexate, hydroxychloroquine, griseofulvin, and sulfasalazine and
3. other treatments with unproven efficacy (eg, mycophenolate mofetil)

## ► Lichen planus of the oral mucosa:

1. Topical steroids;
2. Topical calcineurin inhibitors;
3. Oral or topical retinoids.

## Treatment depends on extent of involvement and the site and morphology of lesions

|                           | <i>Treatment</i>                                              | <i>Comments</i>                                                                 |
|---------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Localized LP</b>       | Topical steroids (medium potency)<br>+<br>Oral antihistamines | Flattens lesions<br>Reduces itching                                             |
| <b>Extensive LP</b>       | PUVA<br>Oral steroids<br>Acitretin                            | Try weekly steroids (called oral mini pulse-OMP) in resistant cases             |
| <b>Hypertrophic LP</b>    | Potent topical steroids + salicylic acid<br>Antihistamines    | Response better, if used underocclusion*.<br>Intralesional steroids             |
| <b>LP of nails, scalp</b> | Oral steroids (can use OMP)                                   | Watch for side effects of steroids!                                             |
| <b>Erosive mucosal LP</b> | Dapsone + oral steroids (can use OMP)<br>Acitretin            | Very effective, especially if lesions ulcerated but requires careful monitoring |

# *Morbidity:*

- ▶ In lichen planus, atrophy and scarring are seen in hypertrophic lesions and in lesions on the scalp.
- ▶ Cutaneous lichen planus does not carry a risk of skin cancer, but ulcerative lesions in the mouth, particularly in men, do have a higher incidence of malignant transformation.
- ▶ However, the malignant transformation rate of oral lichen planus is low.
- ▶ Vulvar lesion in women may also be associated with squamous cell carcinoma.

# Complication:

- ▶ Malignant transformation has been reported in ulcerative oral lichen planus (OLP).
- ▶ Cutaneous hypertrophic lichen planus resulting in squamous cell carcinoma (SCC).
- ▶ Pruritic and painful vulvar lichen planus has been a precursor to SCC.
- ▶ Infection, osteoporosis, adrenal insufficiency, bone marrow suppression, renal damage, hyperlipidemia, and growth restriction in children may occur due to medication adverse effects.
- ▶ Postinflammatory/residual hyperpigmentation may be a common marker after lichen planus has subsided.
- ▶ Alopecia associated with lichen planus is often permanent.
- ▶ Hepatitis C virus infection may be present in 16% of lichen planus patients. Additionally, a meta-analysis reported that hepatitis C virus seropositivity could be 6 times higher in oral lichen planus



THANK YOU